

AI BASED SENTIMENT AND EMOTION ANALYSIS FOR YOUTUBE COMMENTS

A PROJECT REPORT

Submitted by

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I hereby declare that the work entitled “ **AI BASED SENTIMENT AND EMOTION ANALYSIS FOR YOUTUBE COMMENTS** ” is submitted in partial fulfillment of the requirement for the award of the degree in B.E, in University College of engineering, BIT Campus, Anna University, Tiruchirappalli. It is the record of my own work carried out during the academic year 2025 – 2026 under the supervision and guidance of **Dr.K.AMBIKA**, Assistant Professor, Department of Computer Science and Engineering, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli. The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has not been submitted for the award of any other degree, either in this or any other university.

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ABSTRACT

The rapid growth of social media platforms like YouTube has resulted in a massive volume of user-generated comments, especially on technology product reviews such as smartphones, laptops, and gadgets. These comments contain valuable insights about user opinions, experiences, and expectations, but due to their unstructured nature, extracting meaningful information manually becomes difficult and time-consuming. This project presents a system for analyzing sentiment and emotion in YouTube comments using Natural Language Processing (NLP) and deep learning techniques. The proposed system collects comments either through user input or the YouTube Data API and processes them through a structured pipeline that includes preprocessing, sentiment analysis, and emotion detection. Multiple models such as TextBlob, VADER, BERT, and RoBERTa are used to perform sentiment classification, and their outputs are combined using an ensemble voting mechanism to improve accuracy. In addition, a deep learning model based on RoBERTa, BiLSTM, and attention mechanism is used to detect emotions such as joy, anger, sadness, and neutral. The system provides results through an interactive Streamlit interface along with visualizations for better understanding. Performance evaluation shows that the ensemble approach achieves higher accuracy compared to individual models. The system is capable of handling mixed sentiments and informal language, making it suitable for real-world applications. Overall, the proposed system offers an efficient and scalable solution for analyzing user feedback and supports better decision-making in technology product evaluation.

சுருக்கம்

சமூக ஊடக தளங்களான YouTube போன்றவை வேகமாக வளர்ந்து வருவதால், குறிப்பாக ஸ்மார்ட்போன், லேப்டாப் மற்றும் பிற தொழில்நுட்ப பொருட்கள் தொடர்பான பயனர் கருத்துக்கள் மிக அதிகமாக உருவாகின்றன. இந்த கருத்துக்கள் பயனர் அனுபவம், விருப்பம் மற்றும் எதிர்பார்ப்புகள் குறித்து முக்கியமான தகவல்களை கொண்டுள்ளன. ஆனால், இவை அமைப்பற்ற (unstructured) வடிவில் இருப்பதால் அவற்றை கையால் பகுப்பாய்வு செய்வது கடினமும் நேரம் பிடிப்பதுமானதாகும். இந்த திட்டம் இயற்கை மொழி செயலாக்கம் (NLP) மற்றும் ஆழ்ந்த கற்றல் (Deep Learning) நுட்பங்களை பயன்படுத்தி YouTube கருத்துக்களில் உள்ள உணர்ச்சி (sentiment) மற்றும் உணர்வுகளை (emotion) பகுப்பாய்வு செய்யும் ஒரு அமைப்பை முன்வைக்கிறது. இந்த அமைப்பு பயனர் உள்ளீடு அல்லது YouTube Data API மூலம் கருத்துக்களை சேகரித்து, அவற்றை சுத்திகரிப்பு, உணர்ச்சி பகுப்பாய்வு மற்றும் உணர்வு கண்டறிதல் ஆகிய கட்டங்களின் மூலம் செயலாக்குகிறது. TextBlob, VADER, BERT மற்றும் RoBERTa போன்ற பல மாதிரிகள் பயன்படுத்தப்பட்டு உணர்ச்சி வகைப்படுத்தல் செய்யப்படுகிறது மற்றும் அவற்றின் முடிவுகள் ensemble voting முறையின் மூலம் இணைக்கப்பட்டு அதிக துல்லியத்தை பெறுகிறது. மேலும், RoBERTa, BiLSTM மற்றும் attention முறைகளை பயன்படுத்தி மகிழ்ச்சி, கோபம், துக்கம் போன்ற உணர்வுகளையும் கண்டறிகிறது. முடிவுகள் Streamlit அடிப்படையிலான பயனர் இடைமுகத்தின் மூலம் காட்சிப்படுத்தப்படுகின்றன. இந்த அமைப்பு தனிப்பட்ட மாதிரிகளை விட அதிக துல்லியத்தையும் நம்பகத்தன்மையையும் வழங்குகிறது மற்றும் கலப்பு கருத்துக்களையும் சரியாக புரிந்து கொள்ளும் திறன் கொண்டது. இந்த திட்டம் தொழில்நுட்ப பொருட்கள் தொடர்பான பயனர் கருத்துக்களை சிறப்பாக புரிந்து கொள்ள உதவும் ஒரு பயனுள்ள தீர்வாகும்.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CSV	Comma-Separated Values
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
POS	Part of Speech
BERT	Bidirectional Encoder Representations from Transformers
VADER	Valence Aware Dictionary and sEntiment Reasoner
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
NRClex	NRC Emotion Lexicon
BERT	Bidirectional Encoder Representations from Transformers
TF-IDF	Term Frequency-Inverse Document Frequency
SIA	Sentiment Intensity Analyzer (used by VADER)
GloVe	Global Vectors for Word Representation
PCA	Principal Component Analysis
PyTorch	Python-based Scientific Computing Framework
RoBERTa	Robustly optimized BERT approach
UMAP	Uniform Manifold Approximation and Projection
URL	Uniform Resource Allocator
IO	Input Output
ML Pipeline	Machine Learning Pipeline
UI	User Interface

CHAPTER 1

INTRODUCTION

In the age of digital communication, YouTube has emerged as a major platform where users openly express their opinions and emotions. Although these comments are unstructured in nature, they hold significant potential for understanding public sentiment and emotional behavior. This project focuses on designing an intelligent system that can automatically interpret and analyze such feedback using techniques from Natural Language Processing (NLP) and Deep Learning.

The workflow begins with extracting video comments and metadata via the YouTube Data API. The raw textual data undergoes thorough preprocessing, which includes cleaning, normalization, and transformation to ensure it is suitable for analysis. Sentiment classification is carried out using a combination of models—TextBlob, VADER, BERT, and RoBERTa—enabling both rule-based and deep learning-based interpretations. A majority-voting mechanism is used to consolidate sentiment predictions into a final label.

To capture emotional nuances, the system employs a deep neural network architecture that integrates RoBERTa embeddings, a bidirectional LSTM layer, and an attention module. This allows for multi-emotion tagging within individual comments. All processed insights are delivered through a visual interface built using Streamlit. This makes it a valuable tool for researchers, analysts, and decision-makers seeking to monitor public perception.

1.1 BACKGROUND

1.1.1 CONTEXT OF THE PROBLEM

In today's digital era, YouTube has become a major platform where users share their opinions about tech products such as smartphones, laptops, and gadgets through comments. These comments contain valuable feedback, but they are unstructured and large in volume, making manual analysis difficult and time-consuming.

Most existing systems only classify comments into basic sentiments like positive, negative, or neutral. However, real user feedback is often more complex, where a single comment may contain mixed opinions or multiple emotions. Additionally, comments may include informal language or mixed languages, which reduces the accuracy of simple models.

Because of these challenges, there is a need for an automated system that can effectively analyze YouTube tech product review comments and extract meaningful insights.

1.1.2 IMPORTANCE OF THE DOMAIN

Analyzing YouTube comments related to technology products is important because it helps in understanding real user experiences, product performance, and overall customer satisfaction. Users often share honest opinions about features such as camera quality, battery life, performance, and usability, which makes these comments a valuable source of feedback.

Companies and content creators can use this information to identify strengths and weaknesses in their products and make improvements based on actual user needs.

In addition, this domain provides significant benefits for researchers, analysts, and businesses by offering insights into user preferences, behavior patterns, and emerging trends in technology products. By studying large volumes of comments, it becomes possible to identify common issues, popular features, and changing customer expectations over time. This helps organizations make data-driven decisions and improve their strategies.

The use of Natural Language Processing (NLP) and Deep Learning techniques further enhances the efficiency of analyzing such large-scale unstructured data. These techniques enable automated processing of comments, reducing manual effort and improving accuracy. Moreover, combining sentiment analysis with emotion detection provides a deeper understanding of user opinions, as it captures not only whether feedback is positive or negative but also the underlying emotions behind it. This leads to more meaningful insights and supports better product development, marketing strategies, and customer engagement.

1.2 PROBLEM STATEMENT

YouTube contains a huge number of comments related to tech product reviews such as mobiles, laptops, and gadgets. These comments carry valuable user opinions, but most of them remain unprocessed and unanalyzed. Since the data is unstructured and continuously increasing, it is difficult to manually understand user sentiment and emotions.

Another major issue is that user comments are not always simple. A single comment can express multiple opinions or emotions at the same time, making it challenging to analyze accurately using basic techniques. Therefore, there is a need for an intelligent system that can automatically collect, process, and analyze YouTube comments to extract meaningful sentiment and emotional insights.

1.3 OBJECTIVES

The primary objective of this project is to develop an intelligent and efficient system for analyzing YouTube comments related to technology product reviews using sentiment analysis and emotion detection techniques. The system aims to extract meaningful insights from user-generated content by identifying whether a comment expresses a positive, negative, or neutral sentiment, while also detecting underlying emotions such as joy, anger, and sadness. To achieve this, the project integrates multiple Natural Language Processing and deep learning models, including TextBlob, VADER, BERT, and RoBERTa, along with an ensemble approach to improve accuracy and reliability. The system is designed to handle unstructured and informal text commonly found in YouTube comments, including mixed sentiments within a single comment. In addition, the results are presented through an interactive interface, enabling users to easily understand and interpret the analysis.

The next objective of this project is to collect YouTube comments and related information using the YouTube Data API and preprocess the raw data by performing normalization, tokenization, and removal of special characters to ensure data consistency. Another objective is to apply multiple sentiment analysis models to classify comments into positive, negative, or neutral categories, thereby improving prediction accuracy through comparison and ensemble techniques.

Additionally, the project aims to detect a wide range of emotions using a deep learning model based on RoBERTa, BiLSTM, and attention mechanisms for deeper insight into user opinions. Further, the system focuses on generating visualizations such as sentiment distribution and emotion classification charts to enhance result interpretation. Finally, the project aims to develop an interactive Streamlit-based web application that allows users to input comments and view analysis results in a clear and user-friendly dashboard.

CHAPTER 2

LITERATURE SURVEY

The literature review provides an overview of existing research and techniques related to sentiment and emotion analysis of YouTube comments. Various studies have explored different machine learning and deep learning approaches to understand user opinions expressed on social media platforms. These works highlight the importance of analyzing public feedback for applications such as product evaluation, marketing strategies, and trend analysis.

Previous research has used methods like Support Vector Machines, Naïve Bayes, and lexicon-based approaches such as VADER and TextBlob for sentiment classification. More recent studies have focused on advanced deep learning models like BERT, RoBERTa, CNN, and LSTM to improve accuracy and capture contextual meaning. Some works also combine multiple models or include additional features like metadata and time-series analysis to enhance performance.

However, despite these advancements, many existing approaches face limitations such as lack of emotion detection, difficulty in handling multilingual data, high computational cost, and limited real-time processing capability.

This project builds upon these existing studies by proposing a system that integrates multiple sentiment models with deep learning-based emotion detection, providing a more accurate and comprehensive analysis of YouTube tech product review comments.

2.1 REVIEW OF EXISTING SYSTEMS

Paper 1: Zoti et al - Sentiment Analysis of YouTube Comments Using Machine Learning Models (2025)

This research compares various machine learning models such as Random Forest, Naïve Bayes, and SVM for sentiment analysis of YouTube comments. The study finds that Random Forest achieves relatively high accuracy due to its ability to handle non-linear relationships in data. The paper also discusses preprocessing techniques such as tokenization and stop-word removal. However, the focus is mainly on traditional models without incorporating deep learning techniques.

Limitation: Does not include emotion detection

Paper 2: Deroy et al - Sarcasm Detection in Multilingual YouTube Comments Using LLMs (2024)

This paper focuses on detecting sarcasm in multilingual YouTube comments using large language models (LLMs). The study highlights that sarcasm significantly affects sentiment

classification accuracy, as sarcastic comments often contain contradictory meanings. The proposed model uses contextual embeddings and fine-tuning techniques to identify sarcasm patterns in mixed-language comments. Experimental results show improvement compared to traditional models, especially in detecting sarcastic expressions.

Limitation: The model struggles with mixed sentiments and cannot combine multiple models for better accuracy.

Paper 3: AS et al - Machine Learning Based Sentiment Analysis of Informal YouTube Comments (2024)

This study applies machine learning techniques such as Naïve Bayes and SVM to analyze sentiment in informal YouTube comments. It highlights the challenges of handling slang, abbreviations, and unstructured text commonly found in user-generated content. The system performs basic preprocessing and classification but lacks deeper contextual understanding.

Limitation: Poor handling of informal language and lack of deep learning models.

Paper 4: : Bashiri et al – Transformer Based Comparative Study for Sentiment Analysis (2024)

This paper compares various transformer models such as BERT, RoBERTa, and T5 for sentiment classification tasks. The study shows that transformer models significantly outperform traditional approaches in terms of accuracy and contextual understanding. However, it mainly focuses on evaluating models individually rather than combining them.

Limitation: Uses single-model approach without leveraging ensemble techniques.

Paper 5: IRJAET Study - Multilingual YouTube Comment Sentiment Analysis Using XLM RoBERTa (2025)

This paper introduces a multilingual sentiment analysis system using XLM-RoBERTa. It highlights the importance of handling diverse languages in social media data. The model performs well in multilingual scenarios but requires complex setup and lacks user interaction features.

Limitation: No user-friendly interface for analysis.

Paper 6: Khan et al – RNN Based Emotion Detection in Text (2023)

This research uses Recurrent Neural Networks (RNN) for detecting emotions in textual data. The model captures sequential relationships in sentences and identifies basic emotional categories. However, it struggles with multiple emotions in a single comment and lacks integration with sentiment analysis systems.

Limitation: Low accuracy for multi-emotion detection and no integration with sentiment analysis.

2.2 RESEARCH GAP

From the review of existing literature, it is observed that many research works have focused on sentiment analysis of YouTube comments using machine learning and deep learning techniques. Traditional models such as Naïve Bayes, Support Vector Machine, and lexicon-based approaches like TextBlob and VADER provide moderate accuracy but fail to capture contextual meaning and mixed sentiments present in user comments. Recent studies using transformer-based models such as BERT and RoBERTa have improved accuracy; however, these models are computationally expensive and are often used individually without combining their strengths.

Another major limitation identified is that most existing systems focus only on sentiment classification (positive, negative, neutral) and do not perform emotion detection, which is essential for understanding deeper user opinions. Additionally, many systems are not capable of handling multilingual or informal text commonly found in YouTube comments. Handling sarcasm and mixed sentiment within a single comment also remains a challenge in existing approaches.

Furthermore, most implementations lack an integrated system that combines data collection, preprocessing, multi-model analysis, and visualization in a single platform. Many studies do not provide user-friendly interfaces, making it difficult for non-technical users to interpret results. Real-time analysis and scalability are also limited in several existing systems.

To address these gaps, the proposed system integrates multiple sentiment models using an ensemble approach to improve accuracy, incorporates deep learning-based emotion detection for deeper insights, and provides an interactive Streamlit interface for easy visualization and analysis of YouTube tech product review comments.

2.3 COMPARATIVE ANALYSIS

Various existing systems for sentiment and emotion analysis of YouTube comments have been studied to understand the different approaches used in this domain. These systems mainly rely on traditional machine learning models, lexicon-based techniques, and deep learning methods. Machine learning models such as Naïve Bayes and Support Vector Machine provide moderate accuracy but are limited in capturing contextual meaning in complex and informal text. Transformer-based models like BERT and RoBERTa offer improved performance by understanding contextual relationships between words; however, most implementations use these models individually without combining their strengths.

Some research works focus on multilingual sentiment analysis and sarcasm detection, which enhance the understanding of user-generated content. Despite these advancements, handling mixed sentiments within a single comment remains a challenge. In addition, many systems concentrate only on sentiment classification and do not include emotion detection, which is important for gaining deeper insights into user opinions. Another limitation is the absence of user-friendly interfaces, making it difficult for non-technical users to interact with the system.

To overcome these limitations, the proposed system integrates multiple models using an ensemble approach, incorporates emotion detection, and provides an interactive interface for effective analysis and visualization. The comparison of different methods, tools, performance, and limitations is presented in the following table.

TABLE 1: COMPARITIVE ANALYSIS

Method	Tools/ Models	Accuracy/ Performance	Key Findings	Proposed System Improvement
Machine Learning Approach	Random Forest, Naïve Bayes, SVM	Moderate to High (RF performs better)	Random Forest handles non-linear data effectively and improves sentiment prediction	Integrate emotion detection and combine models using ensemble approach
Deep Learning (Sarcasm Detection)	LLMs, Contextual Embeddings	Improved over traditional models	Detects sarcasm in multilingual comments and improves understanding	Handle mixed sentiment using ensemble models and improve contextual analysis
Traditional ML Classification	Naïve Bayes, SVM	Moderate	Works well for basic classification with preprocessing	Use transformer models (BERT, RoBERTa) for better contextual understanding

Method	Tools/ Models	Accuracy/ Performance	Key Findings	Proposed System Improvement
Transformer- Based Model	BERT, RoBERTa, T5	High	Provides strong contextual understanding and higher accuracy	Combine multiple transformer models using ensemble voting for better performance
Multilingual Deep Learning	XLM- RoBERTa	High (for multilingual data)	Supports sentiment analysis across different languages	Provide user- friendly interface and support interactive visualization
Sequential Deep Learning	RNN	Moderate	Captures sequential relationships and basic emotion patterns	Improve multi- emotion detection and integrate with sentiment analysis system

CHAPTER 3

PROPOSED SYSTEM

3.1 EXISTING WORK

Existing systems for YouTube comment analysis mainly focus on identifying sentiment using single-model approaches. In these systems, comments related to technology product reviews such as smartphones, laptops, and gadgets are collected using the YouTube Data API. The collected comments are preprocessed by removing special characters, converting text to lowercase, and filtering unnecessary words. After preprocessing, traditional sentiment analysis techniques such as TextBlob or VADER are applied to classify comments into positive, negative, or neutral categories.

Some existing systems also use machine learning models such as Support Vector Machine (SVM) and Naïve Bayes for sentiment classification. While these models provide basic sentiment prediction, they often fail to capture contextual meaning and mixed opinions present in technology product review comments. For example, a comment like "Camera is good but battery is poor" contains both positive and negative sentiments, which basic models cannot classify accurately.

In recent works, deep learning models such as BERT have been used to improve sentiment classification. These models understand contextual relationships between words and provide better performance than lexicon-based approaches. However, most existing systems rely on a single model and do not combine multiple models for improved accuracy. Additionally, many systems focus only on sentiment classification and do not include emotion detection.

Another limitation of existing systems is the lack of interactive user interfaces. Most implementations are offline scripts that generate textual outputs without providing user-friendly visualization. Moreover, multilingual comments and informal language commonly found in tech product reviews are not handled effectively.

These limitations highlight the need for an improved system that combines multiple sentiment models, includes emotion detection, supports different types of comments, and provides an interactive interface for analyzing YouTube tech product review comments.

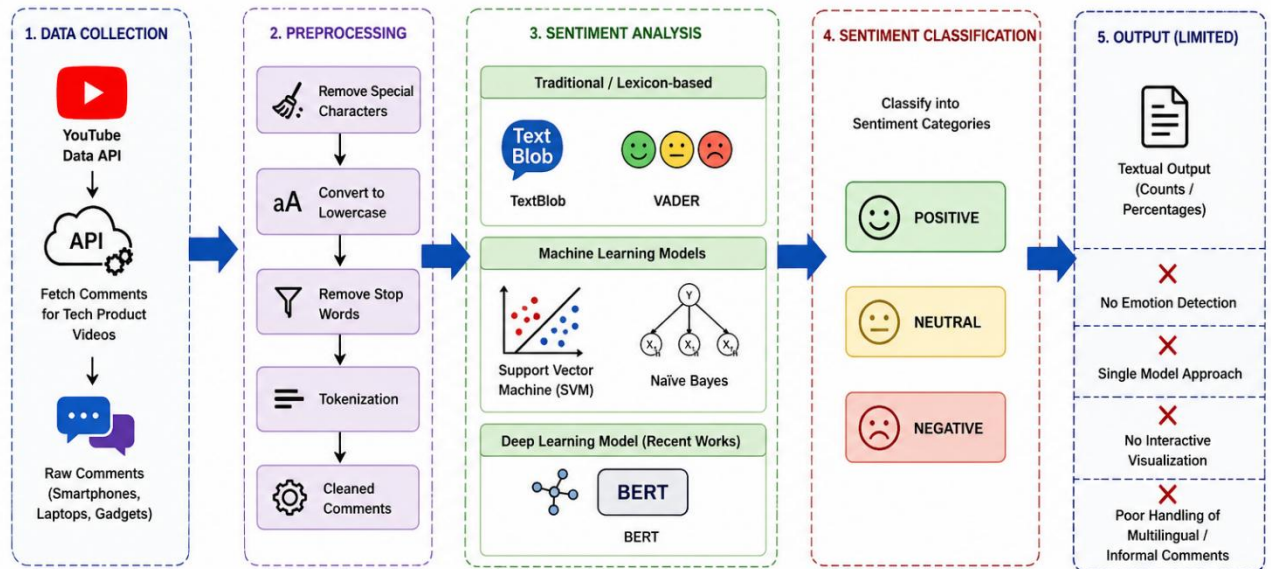


FIGURE 1: EXISTING SYSTEM ARCHITECTURE

3.1.1 DRAWBACKS

Limited Sentiment Precision: Traditional approaches such as VADER and TextBlob often misinterpret sentiment due to their inability to handle context, irony, or layered emotional cues, which results in suboptimal accuracy.

Absence of Emotion Categorization: Many systems only tag sentiments as positive, negative, or neutral, without detecting specific emotional states like sadness, happiness, fear, or anger, which are essential for in-depth analysis.

Lack of Advanced Language Models: These systems do not utilize cutting-edge deep learning architectures like BERT or RoBERTa, which are more effective in capturing linguistic subtleties and contextual nuances.

Single-Model Dependency: Current solutions depend on just one classifier, lacking ensemble-based strategies that can combine predictions from multiple models to improve robustness and reduce errors.

No Interactive or Deployable Interface: These systems are typically not integrated into user-friendly platforms like Streamlit, nor are they made accessible as deployable web applications.

Lack of Ground Truth Validation: Most models do not validate their predictions against manually annotated data, which compromises the ability to confirm their accuracy and trustworthiness.

3.2 PROPOSED SYSTEM

The project titled "AI Based Sentiment and Emotion Analysis For Youtube Comments" is designed to develop a robust and integrated system for examining user opinions expressed in YouTube comments. It begins by using the YouTube Data API to efficiently extract essential data, including video details, user comments, likes, and reply threads. Once data collection is complete, a rigorous preprocessing pipeline is applied. This involves breaking down the text into tokens, removing irrelevant words (stopwords), performing lemmatization to reduce words to their root forms, and normalizing the content for uniformity—ensuring the dataset is clean and well-prepared for analysis.

For sentiment classification, the system employs multiple natural language processing models—TextBlob, VADER, BERT, and RoBERTa—combined through an ensemble strategy. This approach improves the accuracy and consistency of sentiment labeling by resolving discrepancies among model outputs.

In terms of emotion recognition, the framework integrates NRCLEX and a tailored deep learning architecture that blends RoBERTa, BiLSTM, and attention mechanisms. This enables the model to accurately detect and classify emotions like happiness, anger, fear, trust, and sadness—even in ambiguous or context-heavy comments.

A dedicated trend visualization module charts how public sentiment and emotional expressions change evolving discussion patterns. Finally, the entire solution is deployed as a real-time, user-friendly web application built with Streamlit, making it accessible to researchers, analysts, and the general public who wish to explore public reactions to global events through YouTube data.

3.2.1 FEATURES OF PROPOSED SYSTEM

High Sentiment Accuracy: Combines multiple models (TextBlob, VADER, BERT, RoBERTa) with ensemble voting to improve sentiment prediction reliability.

Emotion Detection Capability: Supports multi-label emotion classification (e.g., joy, fear, anger) using deep learning models like RoBERTa + BiLSTM + Attention.

User-Friendly Deployment: Offers an interactive Streamlit-based web application for real-time analysis and visualization.

Support for Large Datasets: Designed to handle large volumes of YouTube comments efficiently through batch processing.

Visual Insights: Provides visualizations like pie charts, and bar graphs for better understanding of trends.

Scalable and Modular: Each module (sentiment, emotion) can be independently improved or extended for future research.

3.3 SYSTEM ARCHITECTURE

The diagram shows a process for analyzing data. It starts with gathering data, then cleaning it to remove any unnecessary parts. After that, important features are picked out. The main analysis includes three parts: finding main topics, understanding emotions, and determining overall feelings (positive, negative, or neutral). Once the data is analyzed, the findings are checked for accuracy and then turned into easy-to-understand visuals. Finally, these insights are presented in a user-friendly way for better understanding. This process helps turn raw data into useful information.

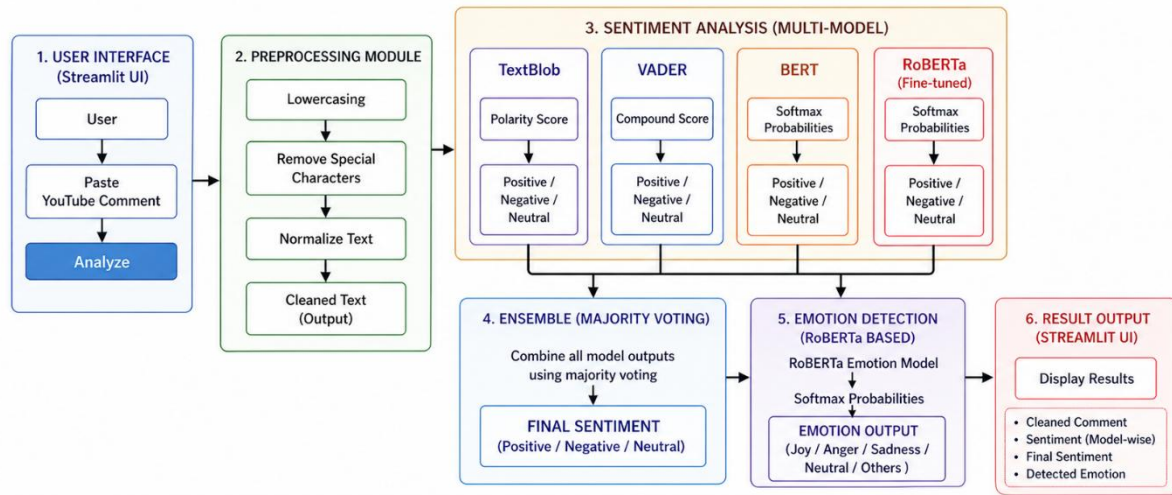


FIGURE 2: SYSTEM ARCHITECTURE

Input Layer

The input layer is responsible for collecting user data for analysis. In this system, the input consists of YouTube tech product review comments, which can either be entered manually by the user through the interface or fetched automatically using the YouTube Data API. Along with the comments, additional metadata such as likes and reply counts may also be collected. This layer ensures that relevant and real-time data is available for further processing.

Preprocessing Layer

The preprocessing layer focuses on cleaning and preparing the raw text data. Since YouTube comments are often unstructured and may contain noise such as special characters, emojis, or inconsistent formatting, this layer standardizes the data. The text is converted into lowercase, unnecessary symbols are removed, and stopwords are filtered out. Tokenization is also performed to break the text into meaningful units. This step ensures that the data becomes consistent and suitable for accurate analysis.

Embedding Layer

The embedding layer converts textual data into numerical representations that can be processed by machine learning models. Advanced transformer-based models such as BERT and RoBERTa generate contextual embeddings that capture the semantic meaning of the text.

This allows the system to understand not just individual words, but also the context in which they are used, improving the overall quality of analysis.

Sentiment Analysis Layer

The sentiment analysis layer applies multiple models to classify the sentiment of the input comment. Models such as TextBlob and VADER use lexicon-based and rule-based approaches, while BERT and RoBERTa provide deep learning-based contextual understanding. Each model independently predicts whether the comment is positive, negative, or neutral, allowing the system to capture different perspectives of sentiment.

Ensemble Layer

The ensemble layer combines the outputs from all sentiment models to produce a final result. Instead of relying on a single model, the system uses a majority voting mechanism to determine the final sentiment. This approach improves accuracy and reliability by reducing the impact of errors from individual models and ensuring a more balanced prediction.

Emotion Detection Layer

The emotion detection layer is designed to identify the emotional tone of the comment. A deep learning architecture combining RoBERTa, BiLSTM, and an attention mechanism is used to analyze the text. This layer is capable of detecting multiple emotions such as joy, anger, sadness, and fear within a single comment, providing deeper insights beyond basic sentiment classification.

Evaluation Layer

In this layer, the outputs generated by different models are evaluated and validated. The system compares predictions from multiple models and checks for consistency. Probability scores are also analyzed to ensure the reliability of the results. This layer helps confirm that the final sentiment and emotion outputs are accurate and trustworthy.

Visualization Layer

The visualization layer presents the results in a clear and understandable format. Sentiment results are displayed using charts such as bar graphs, while emotion results are shown using pie charts. These visual representations help users quickly interpret the analysis and understand patterns in the data without needing technical knowledge.

Output Layer

The output layer delivers the final results to the user through an interactive Streamlit interface. It displays the cleaned comment, final sentiment classification, detected emotions, and model-wise predictions. This layer ensures that all insights are presented in a user-friendly and accessible manner.

System Performance

The overall system is designed to provide accurate and efficient analysis by combining multiple models and deep learning techniques. It reduces manual effort, supports multilingual comments, and delivers real-time results through an interactive dashboard. The integration of ensemble learning and emotion detection makes the system more reliable and effective compared to traditional approaches.

3.4 MODULES AND FUNCTIONALITIES

1. DATA COLLECTION
2. DATA PREPROCESSING
3. SENTIMENT ANALYSIS MODULE
4. EMOTION ANALYSIS MODULE
5. EVALUATION MODULE
6. VISUALIZATION AND TREND ANALYSIS MODULE
7. FRONTEND AND USER INTERFACE MODULE

3.4.1 Data Collection

Data collection starts by configuring access to the YouTube Data API v3 using an API key for authentication. The system retrieves comments, likes, and reply counts from videos. The collected data is then stored in a structured Excel file, enabling easy access for further sentiment and emotion. This automated process ensures the dataset is ready for analysis without manual effort.

3.4.2 Data Preprocessing

The preprocessing module ensures clean and consistent data by removing duplicates and empty comments. It standardizes text through lowercasing, removing special characters and normalizing. Engagement metadata like likes and reply counts is also extracted. The processed data is organized using Pandas, including both raw and cleaned text. Sentiment analysis with TextBlob assigns polarity scores to classify comments as positive, negative, or neutral. An engagement score is also computed, giving more weight to replies to reflect deeper audience interaction.

3.4.3 Sentiment Analysis Module

The sentiment analysis module interprets the emotional tone of YouTube comments using multiple classification models. It begins with data preprocessing to clean and standardize the text. Four models—TextBlob, VADER, BERT, and RoBERTa—are used to assess sentiment, each offering unique strengths in handling informal or contextual language. TextBlob and VADER assign polarity and sentiment labels, while BERT and RoBERTa use deep learning

for more accurate classification. A majority voting method determines the final sentiment label, with an agreement status indicating whether model outputs were consistent or conflicting.

3.4.4 Emotion Analysis Module

The Emotion Analysis module is designed to detect multiple emotions in YouTube comments. It starts by loading relevant data, specifically the text of each comment and its emotion tags. These emotion tags are split into individual labels and transformed into binary vectors using a multi-label encoding approach. The text is then tokenized using the RoBERTa tokenizer, producing inputs suitable for the model. The model architecture combines RoBERTa for contextual embeddings, a bidirectional LSTM for sequence modeling, and an attention layer to emphasize key information in the text. A final dense layer with sigmoid activation enables multi-label emotion prediction. The model is trained using binary cross-entropy loss and optimized with Adam. During evaluation, predicted probabilities are thresholded to assign emotion labels, which are then decoded back into human-readable form.

3.4.5 Evaluation

The evaluation module verifies the performance of sentiment and emotion models used in the system. The project uses libraries such as pandas, numpy, scikit-learn, torch, and transformers for model execution. Each input comment is analyzed using TextBlob, VADER, BERT, and fine-tuned RoBERTa models. The outputs from these models are compared, and the final sentiment is determined using an ensemble majority voting method. The emotion classification is performed using a RoBERTa-based model. The system ensures reliability by checking consistency between model predictions and validating outputs through probability scores. This evaluation confirms that the combined model approach improves sentiment prediction accuracy.

3.4.6 Visualization and Trend Analysis Module

The visualization module displays sentiment and emotion results using simple charts. After analyzing the input comment, the system generates model-wise sentiment outputs and final sentiment results. A chart is used to show sentiment distribution such as positive, negative, and neutral. Another chart displays detected emotion categories. These visual representations help users easily understand the analysis results. The visualization is implemented within the Streamlit interface for interactive display.

3.4.7 Frontend and User Interface Module

The frontend of the system is developed using Streamlit to provide a simple and interactive interface. The user pastes a YouTube tech product review comment into the input box. The system preprocesses the text and displays the cleaned comment. After analysis, the interface shows model-wise sentiment predictions, final sentiment using ensemble voting, and detected emotion. The results are presented clearly using text and chart visualization. The interface also includes loading indicators and error handling to improve user experience.

3.5 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) illustrates the systematic process of tracking public sentiment on YouTube. It begins with data collection from the YouTube API, retrieving video metadata and user comments. The collected data undergoes preprocessing to clean and normalize the text. This refined data is then fed into a sentiment analysis module utilizing models like TextBlob, VADER, BERT, and RoBERTa. Following this, emotion analysis is performed using a combination of RoBERTa and BiLSTM with an attention mechanism. Finally, all processed insights are compiled and presented through an output generator that delivers results via dashboards and Excel exports, enabling real-time understanding of sentiment dynamics.

3.5.1 Level 0 DFD

At the highest level, the system consists of three main entities: the User, the Sentiment and Emotion Analysis System, and the Output Interface. The user provides input in the form of YouTube comments, either manually or through the YouTube Data API. The system processes the input through various stages including preprocessing, sentiment analysis, and emotion detection. The processed results are then displayed to the user through an interactive interface in the form of sentiment classification and emotion insights.

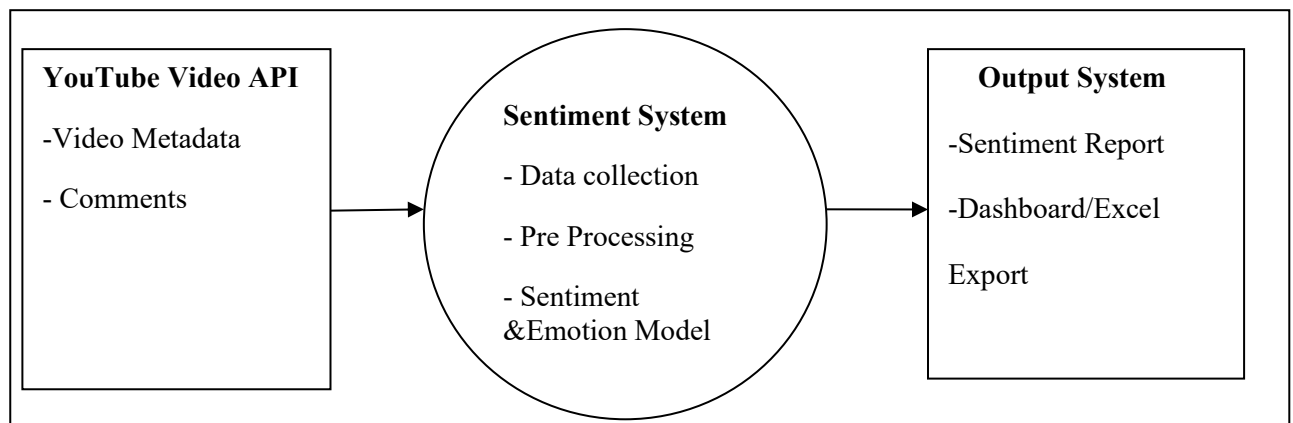


FIGURE 3: DFD Diagram Level 0

3.5.2 Level 1 DFD

At a more detailed level, the system performs a sequence of processing steps to analyze YouTube comments effectively. The process begins with the input of comments either provided directly by the user or retrieved using the YouTube Data API. The collected text data is then preprocessed by cleaning, normalizing, and tokenizing the content to ensure consistency. The processed text is passed through multiple sentiment analysis models, including TextBlob, VADER, BERT, and RoBERTa, where each model independently predicts the sentiment. These outputs are then combined using an ensemble voting mechanism to determine the final sentiment. In parallel, the same preprocessed text is analyzed using a deep learning model to detect emotions such as joy, anger, and sadness. The

system then generates the final output, which includes the overall sentiment, detected emotion, and model-wise predictions. The results are presented through visualizations such as charts and graphs for better understanding. The data involved in this process includes input comments, processed text, and model outputs, which flow sequentially through each stage, ensuring accurate and efficient analysis before delivering the final results to the user.

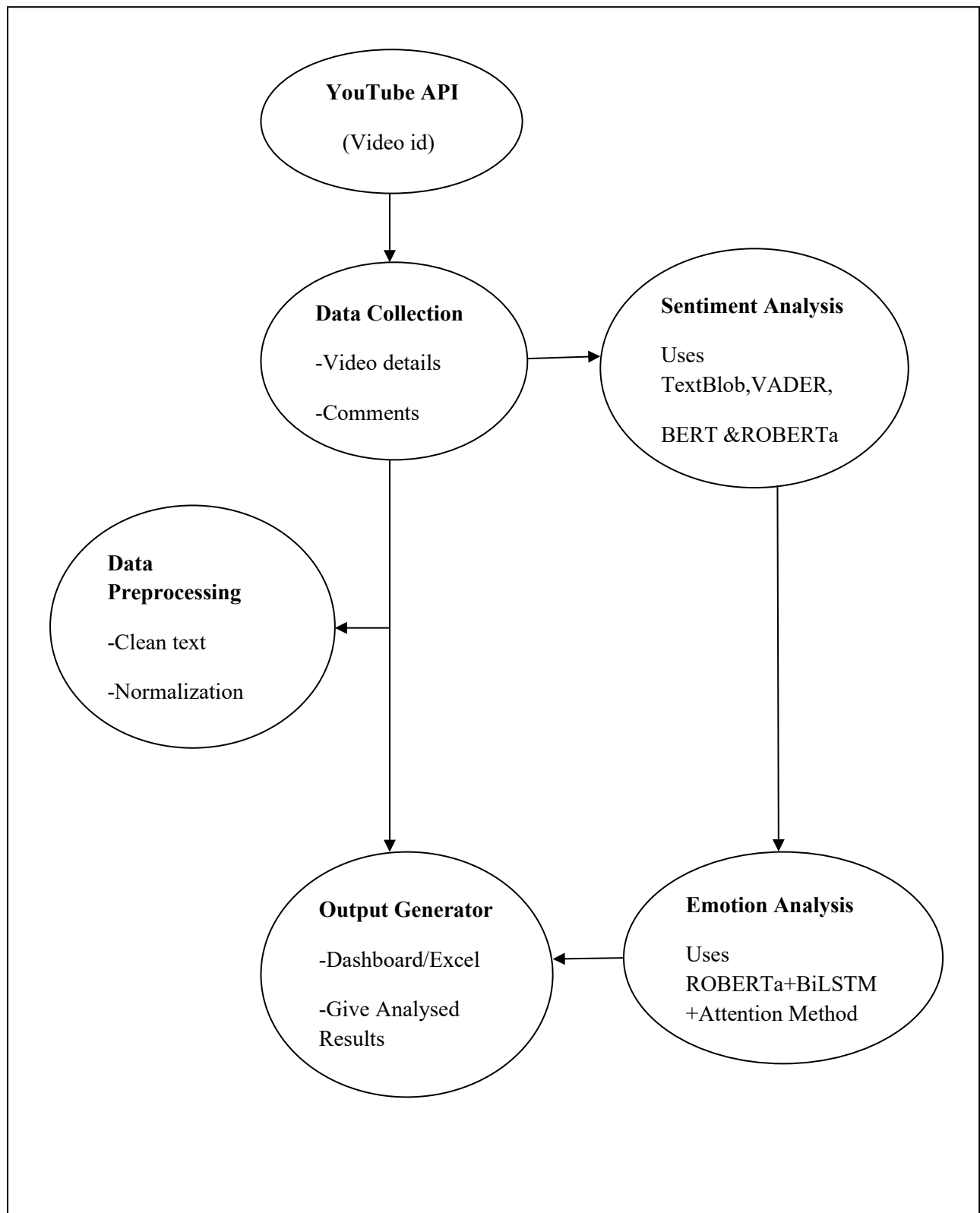


FIGURE 4: DFD Diagram Level 1

3.6 METHODOLOGY

YouTube Data API v3

The data collection process uses the YouTube Data API v3 to retrieve comments from technology product review videos. By providing a video ID, the system collects user comments related to the product. These comments are used as input data for sentiment and emotion analysis. This method enables efficient and structured retrieval of YouTube comments without manual copying.

Preprocessing with NLTK and TextBlob

To prepare the raw comments for analysis, a structured preprocessing workflow is employed using libraries like NLTK and TextBlob. This involves converting all text to lowercase, eliminating unnecessary elements such as emojis, special characters and discarding non-alphabetic content. This preprocessing pipeline ensures that the input text is clean, consistent, and ready for advanced sentiment and emotion analysis.

Sentiment Classification with Ensemble Models

For robust sentiment analysis, the system adopts a hybrid approach that integrates both traditional and deep learning models. TextBlob and VADER perform sentiment scoring based on predefined lexicons, making them effective for analyzing informal language. BERT and RoBERTa, which are transformer-based language models, offer deeper contextual understanding of the text. The final sentiment label is generated through a majority voting mechanism, which selects the most commonly predicted sentiment across all models. This ensemble strategy improves accuracy and stability in handling diverse linguistic inputs.

Emotion Analysis Using RoBERTa + BiLSTM + Attention

To detect nuanced emotional expressions in comments, a deep learning architecture combining RoBERTa embeddings, BiLSTM, and an attention layer is implemented. RoBERTa extracts contextual features from the text, BiLSTM analyzes word sequences in both directions, and the attention mechanism highlights emotionally significant portions of the input. This configuration enables multi-label emotion classification, allowing the model to identify multiple emotions—such as anger, joy, fear, or trust—within a single comment.

User Interface Development with Streamlit

The insights generated from the analysis are presented through a Streamlit-based interactive web application. The dashboard offers real-time visualizations, including sentiment trends, emotion distributions, bar graphs, and word clouds. Users can interact with filters and controls to explore the dataset across different timeframes or categories. The intuitive design ensures accessibility, enabling both technical users and general audiences to gain valuable insights from the analyzed YouTube comments.

3.6.1 Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a structured process used to design, develop, and deploy the YouTube Comment Sentiment and Emotion Analysis system. It ensures that the system is developed in a systematic and efficient manner. The project follows a step-by-step approach from requirement gathering to deployment.

Phase 1 – Requirement Gathering & Analysis

In this phase, the requirements of the system are identified and analyzed. The main objective is to understand the need for analyzing YouTube tech product review comments. Functional requirements include collecting comments, preprocessing text, performing sentiment and emotion analysis, and displaying results through a user interface. Non-functional requirements such as performance, scalability, and usability are also considered.

Phase 2 - System Design & Architecture

In this phase, the overall architecture of the system is designed. The system is divided into different modules such as data collection, preprocessing, sentiment analysis, emotion detection, and visualization. Diagrams like system architecture, data flow diagram, and use case diagram are prepared to represent the system structure clearly.

Phase 3 – Implementation & Coding

During the implementation phase, the system is developed using programming languages and tools. Python is used as the primary language, along with libraries such as TextBlob, VADER, and transformer-based models like BERT and RoBERTa. The user interface is developed using Streamlit. Each module is coded and integrated step by step.

Phase 4 – Testing & Quality Assurance

In this phase, the system is tested to ensure it works correctly. Different test cases are used to verify sentiment and emotion predictions. The performance of models is evaluated using accuracy and consistency checks. Errors are identified and corrected to improve system reliability.

Phase 5 – Deployment & Release

After successful testing, the system is deployed for use. The Streamlit application provides an interactive interface where users can input comments and view results. The system can be hosted locally or using cloud platforms for better accessibility.

Phase 6 – Maintenance & Iteration

The maintenance phase involves updating and improving the system over time. This includes adding new features, improving model accuracy, and handling new types of data such as multilingual comments. Regular updates ensure the system remains efficient and relevant.

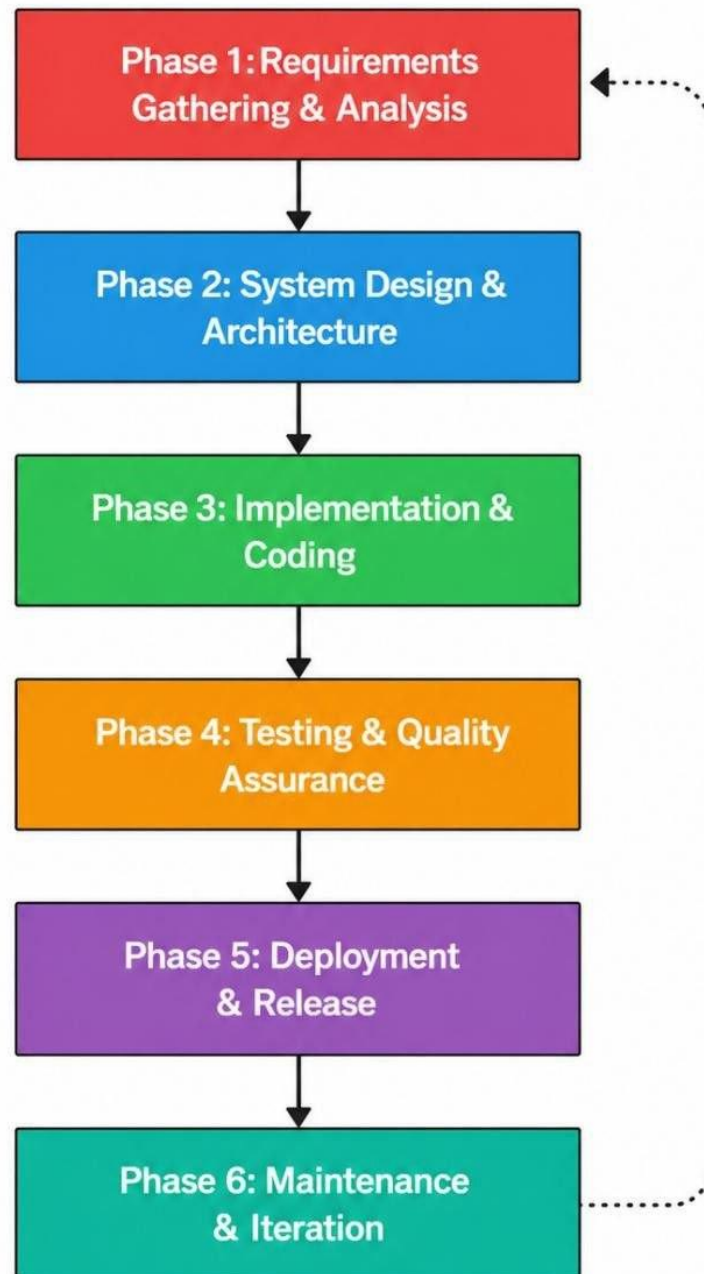


FIGURE 5: SDLC Diagram

3.7.1 FLOWCHART

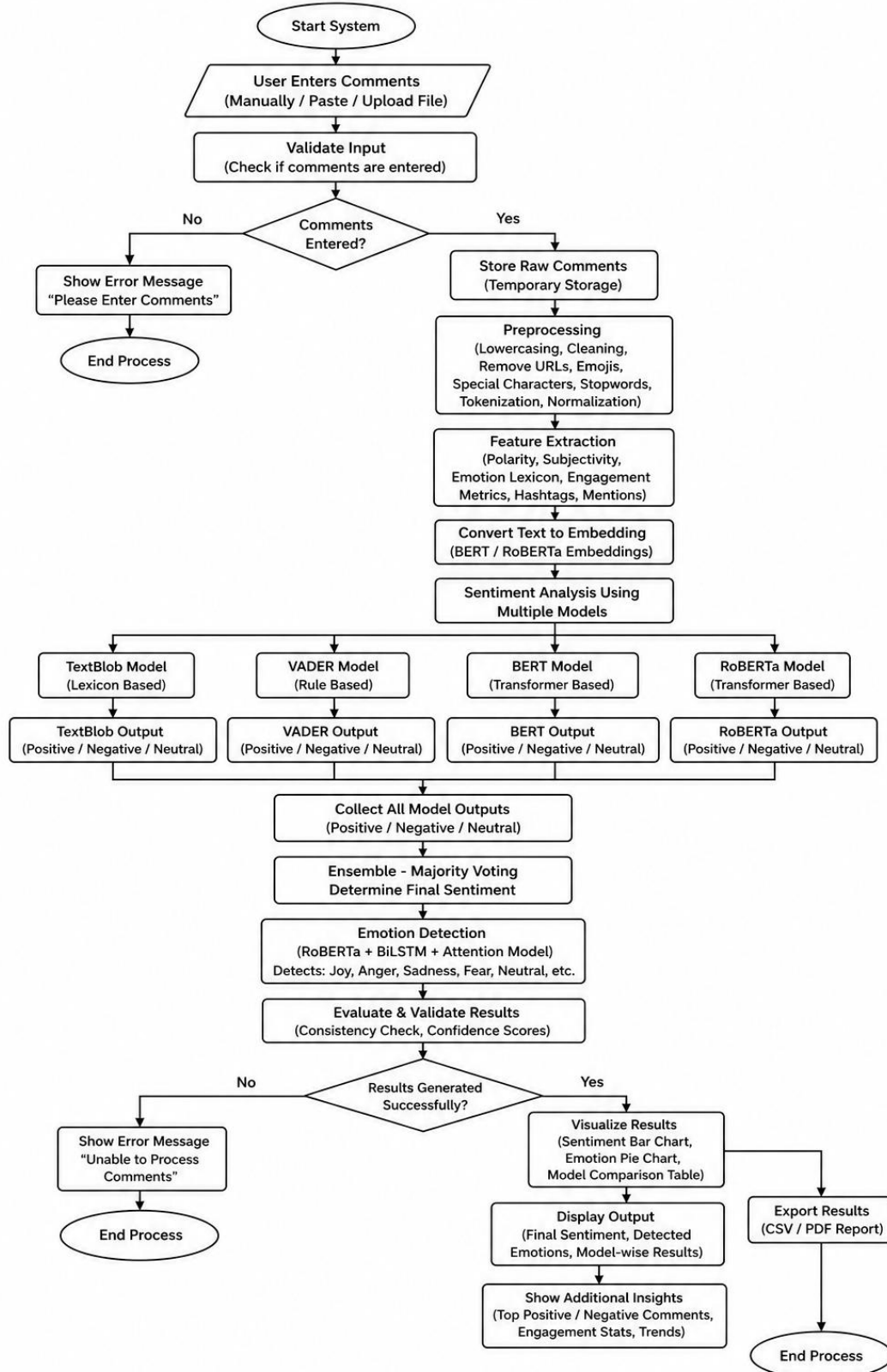


FIGURE 6: FLOWCHART OF YOUTUBE COMMENT SENTIMENT AND EMOTION ANALYSIS SYSTEM

CHAPTER 4

IMPLEMENTATION & RESULTS

4.1 TOOLS & TECHNOLOGIES USED

The project utilizes various tools and technologies for development, data processing, model execution, and visualization. These tools ensure efficient implementation and user-friendly interaction.

TABLE 2: TOOLS & TECHNOLOGIES USED

Category	Tool	Purpose
Programming Language	Python	Core system development
NLP Libraries	TextBlob, VADER	Sentiment analysis
Deep Learning Models	BERT, RoBERTa	Contextual sentiment analysis
Emotion Detection	RoBERTa, BiLSTM, Attention	Emotion classification
Frontend Framework	Streamlit	User interface
Data Source	YouTube Data API	Fetching comments
Visualization	Matplotlib	Graphical representation
IDE	VS Code	Development environment

4.1.1 SOFTWARE REQUIREMENTS

This project utilizes a diverse range of software tools and platforms to support data collection, natural language processing, machine learning, and user-friendly visualization. The environment has been categorized by functionality for clarity and ease of integration.

Programming Environment

Python 3.10: The foundational language for all tasks, from initial data preparation to advanced modeling and final result visualization.

Visual Studio Code (VS Code): A modern integrated development environment (IDE) for writing, debugging, and maintaining Python scripts locally.

Data Processing and Visualization Libraries

Pandas: Essential for structured data manipulation, cleaning, and transformation tasks.

NumPy: Enables efficient numerical operations and matrix processing, which are crucial in data science workflows.

Matplotlib & Seaborn: Libraries used for producing insightful visualizations, including graphs, charts, and statistical plots.

Sentiment and Emotion Analysis

TextBlob & VADER: Lightweight lexicon-based tools designed for analyzing sentiments in short and informal texts such as social media comments.

BERT & RoBERTa: Pre-trained deep learning models that provide context-aware sentiment interpretation based on transformer architecture.

Custom Emotion Model (RoBERTa-BiLSTM-Attention): An advanced multi-label classifier that integrates contextual embeddings with bidirectional LSTM and an attention layer to detect complex emotional patterns.

Web APIs and Communication

YouTube Data API v3: Used to extract structured information such as comments and metadata from YouTube videos.

Requests: A Python HTTP library that enables interaction between the application and web-based APIs or services.

Data Export and Reporting

Openpyxl: Allows the generation and export of structured outputs (e.g., labeled sentiments) into Excel spreadsheets for documentation or further analysis.

Dashboard Deployment and Hosting

Streamlit: A Python-based framework to build responsive and interactive dashboards that display analytical outcomes in real time.

Pyngrok: Facilitates the temporary public hosting of local dashboards by securely exposing them through shareable links.

Development and Testing Utilities

Postman: A popular tool used to test API endpoints and validate request-response flows during the development phase.

Visual Studio Code (VS Code): A modern integrated development environment (IDE) for writing, debugging, and maintaining Python scripts locally.

TABLE 3: SOFTWARE REQUIREMENTS

Category	Software / Tool
Operating System	Windows / Linux / macOS
Programming Language	Python
Frontend	Streamlit
Backend Processing	Python (NLP & ML models)
Libraries	TextBlob, VADER, Transformers (BERT, RoBERTa)
Data Collection	YouTube Data API
Visualization	Matplotlib / Plotly
IDE	VS Code / Google Colab

4.1.2 HARDWARE REQUIREMENTS

The hardware requirements for the proposed system are minimal as the project is developed using lightweight tools such as Python, Streamlit, and pre-trained machine learning models. The system is designed to run efficiently on standard computing devices without the need for high-end infrastructure. A basic system with moderate processing capability is sufficient for development and execution.

A minimum of Intel i3 processor (or equivalent) is required, while an Intel i5 or higher is recommended for smoother performance, especially when running multiple models simultaneously. The system requires at least 4 GB RAM, although 8 GB or more is recommended for better performance during model execution. A minimum of 20 GB free storage is required, and SSD storage is preferred for faster data processing. A stable internet connection is necessary for fetching YouTube comments and downloading required libraries.

TABLE 4: HARDWARE REQUIREMENTS

Component	Minimum Requirement	Recommended
Processor	Intel i3 / AMD equivalent	Intel i5 or higher
RAM	4 GB	8 GB or above
Storage	20 GB HDD	50 GB SSD
Internet	Basic connectivity	High-speed connection

4.2 SYSTEM IMPLEMENTATION

The analysis shows that YouTube tech product review comments express different sentiments such as positive, negative, and neutral. The system successfully classifies comments using multiple models including TextBlob, VADER, BERT, and RoBERTa. The ensemble method improves the final sentiment prediction by combining outputs from all models. The emotion analysis also identifies emotional tone such as joy, anger, sadness, and neutral. The results demonstrate that the system effectively analyzes user opinions in tech product review comments.

4.2.1 BACKEND IMPLEMENTATION

The backend of the proposed system is implemented using Python and structured as a modular pipeline that integrates data processing, sentiment analysis, and emotion detection. The system is organized into multiple functional modules including data collection, preprocessing, sentiment analysis, emotion detection, and ensemble evaluation.

The data collection module uses the YouTube Data API to fetch comments related to tech product review videos. The preprocessing module cleans the raw text by removing special characters, converting text to lowercase, and eliminating unnecessary words to ensure consistency.

The sentiment analysis module integrates multiple models such as TextBlob, VADER, BERT, and RoBERTa, each of which independently predicts sentiment. These predictions are then combined using an ensemble mechanism based on majority voting to improve accuracy.

For emotion detection, a deep learning architecture combining RoBERTa, BiLSTM, and an attention mechanism is used to identify emotions such as joy, anger, sadness, and neutral. The backend processes are executed efficiently using Python libraries and can be run in environments such as Google Colab for handling computational requirements.

4.2.2 FRONTEND IMPLEMENTATION

The frontend of the system is developed using Streamlit, which provides an interactive and user-friendly web interface. The application allows users to input YouTube comments directly into a text field and initiate analysis using a simple button interface.

The interface displays results in a structured format, including the predicted sentiment, detected emotion, and model-wise outputs. Visualization components such as charts are integrated to represent sentiment and emotion distributions.

Streamlit enables real-time interaction, where users can instantly view results without requiring complex setup. The frontend is lightweight, easy to deploy, and can be accessed through a web browser, making the system accessible for both technical and non-technical users.

4.2.3 DATA HANDLING

The system processes textual data rather than structured relational data. The workflow begins with collecting YouTube comments, which are then preprocessed and passed through multiple analysis layers.

The data pipeline ensures smooth flow between modules, where the output of one stage becomes the input for the next. Preprocessed text is transformed into embeddings for deep learning models, enabling contextual understanding.

Intermediate results from different models are stored temporarily during execution, and the final outputs are generated after ensemble evaluation. This design ensures flexibility, scalability, and efficient handling of unstructured text data.

4.2.4 MODEL INTEGRATION AND EXECUTION

The system integrates multiple machine learning and deep learning models to improve performance. Each model processes the input independently, and their outputs are combined to produce a final result.

The execution flow begins with preprocessing, followed by parallel execution of sentiment models. The ensemble module aggregates the predictions and determines the final sentiment. Simultaneously, the emotion detection model processes the same input to identify emotional categories.

The system ensures efficient execution by optimizing preprocessing and model inference steps. The use of pre-trained transformer models reduces training time while maintaining high accuracy.

4.2.5 DEPLOYMENT

The system is deployed as a Streamlit web application, allowing users to access it through a browser. The application can be hosted locally or on cloud platforms such as Streamlit Cloud or similar services.

Deployment ensures that the system is accessible in real-time, providing immediate analysis results. The lightweight nature of the frontend and modular backend design makes the system easy to maintain and update.

4.2.6 YOUTUBE COMMENT SENTIMENT AND EMOTION ANALYSIS

Page Elements:

The user interface includes a title and logo that clearly indicate the purpose of the application, which is to analyze sentiment and emotion in YouTube comments. It provides a large input field where users can easily paste YouTube comments for analysis. An instruction text, “Enter YouTube Comment,” is displayed to guide users on how to use the tool correctly.

Additionally, an analyze button is provided, allowing users to perform sentiment and emotion analysis with a single click on the entered comments.

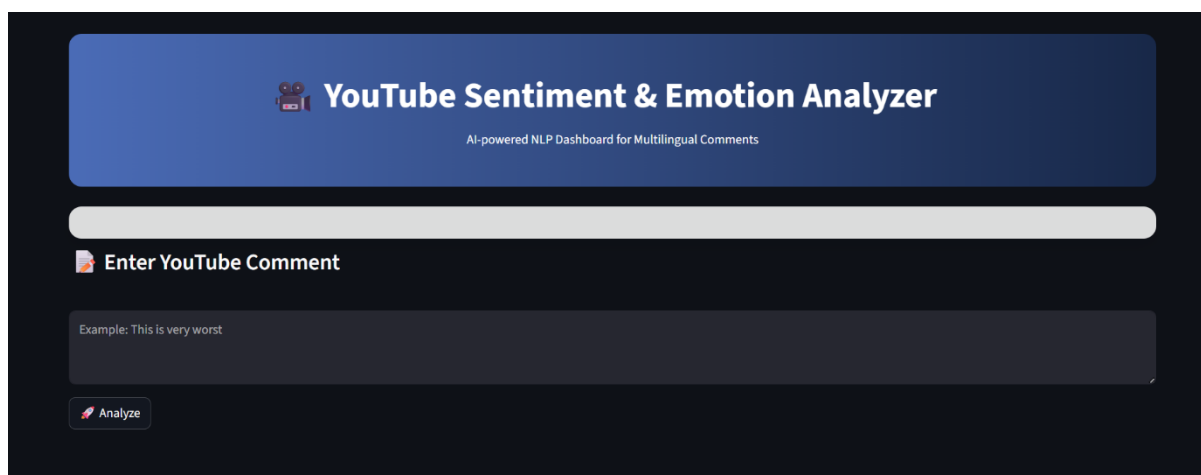


FIGURE 7: FRONT PAGE

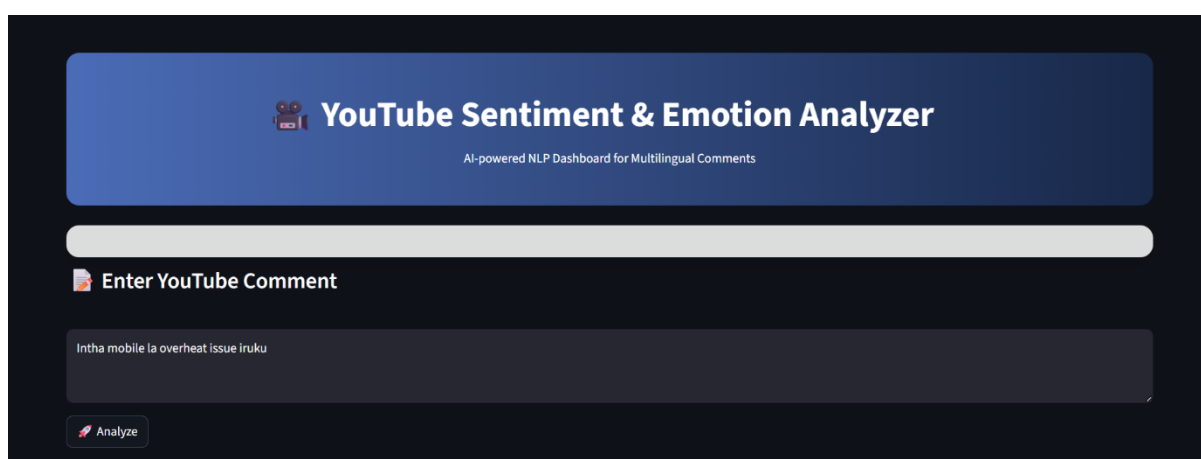


FIGURE 8: PASTE COMMENT

4.3 WORKING MODEL

The working model of the system explains how the input comment is processed step by step to generate sentiment and emotion results.

Step 1: Data Collection

Comments can also be collected automatically using the YouTube Data API by providing a video ID. The collected comments are then stored for further analysis and processing within the system.

Step 2: Input Collection

The user enters a YouTube tech product review comment through the Streamlit interface. This comment acts as the input for the system.

Step 3: Data Preprocessing

The input comment is preprocessed to make it suitable for analysis. This involves converting the text to lowercase, removing special characters and unnecessary words, and performing tokenization. These steps ensure consistency and improve the quality of the data for accurate sentiment and emotion analysis.

Step 4: Sentiment Analysis (Multiple Models)

The cleaned text is then passed to multiple sentiment analysis models, including TextBlob, VADER, BERT, and RoBERTa. Each of these models independently analyzes the input and predicts the sentiment as positive, negative, or neutral.

Step 5: Ensemble Voting

The outputs from all the models are then combined using an ensemble approach. Majority voting is applied to determine the final sentiment, which helps improve overall accuracy and reduces errors from individual model predictions.

Step 6: Emotion Detection

The same preprocessed text is then passed to a deep learning model, specifically a combination of RoBERTa, BiLSTM, and an attention mechanism, which identifies emotions such as joy, anger, sadness, and others.

Step 7: Result Generation

The system generates the final sentiment, the detected emotion, and the individual predictions from each model, providing a complete analysis of the input comment.

Step 8: Visualization

The results are displayed using visualizations such as sentiment distribution and emotion distribution charts, which help users easily understand and interpret the output.

Step 9: Display in User Interface

All outputs are presented through a Streamlit dashboard, where users can clearly view insights and results in an interactive and easy-to-understand format.

4.4 RESULTS & OUTPUTS

4.4.1 ANALYSIS RESULTS

The entered comment “Intha mobile la overheat issue irukku” is analyzed by the system, which identifies the sentiment as negative, indicating a concern or dissatisfaction expressed by the user. Additionally, the system detects the underlying emotion as anger, as the comment reflects frustration regarding the overheating issue in the mobile device.

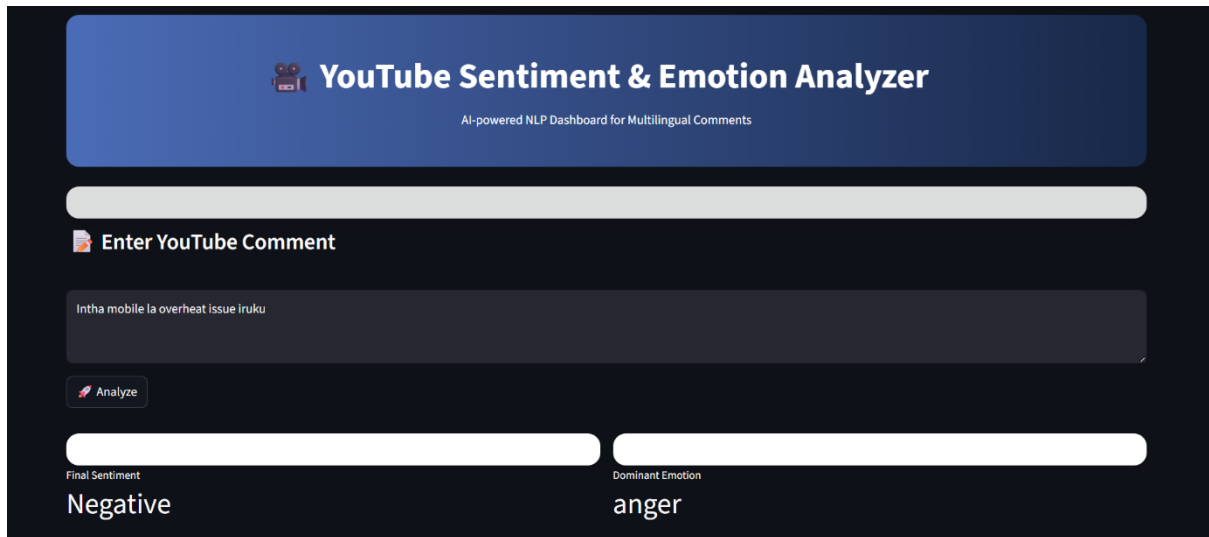


FIGURE 9: ANALYSIS RESULT

4.4.2 SENTIMENT DISTRIBUTION

All the data points analyzed in this session are classified under the “Negative” sentiment category, indicating that the comments entered reflect dissatisfaction or concern. There are no instances of “Positive” or “Neutral” sentiments observed in this session. In the visualization, the y-axis represents the proportion of analyzed comments ranging from 0.0 to 1.0, while the x-axis displays the sentiment categories. However, only the “Negative” category appears due to the absence of the other sentiment classes.

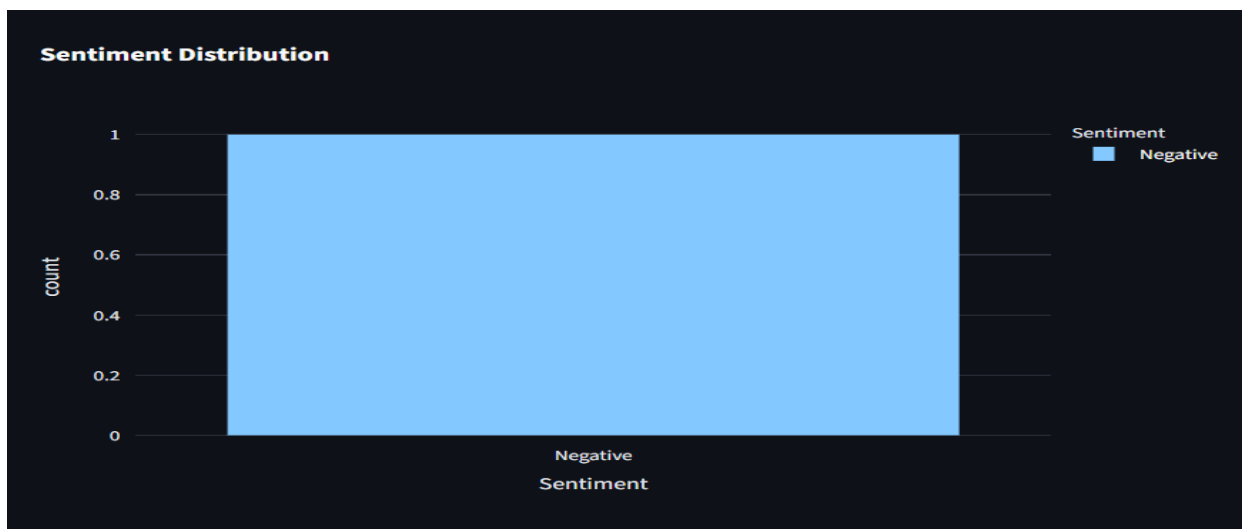


FIGURE 10: SENTIMENT ANALYSIS

4.4.3 EMOTION DISTRIBUTION

The pie chart consists of a single slice representing the emotion category “Anger.” The entire chart is filled, indicating a value of 1.0 (100%), which means that all analyzed comments in this session are classified under the emotion “Anger.”

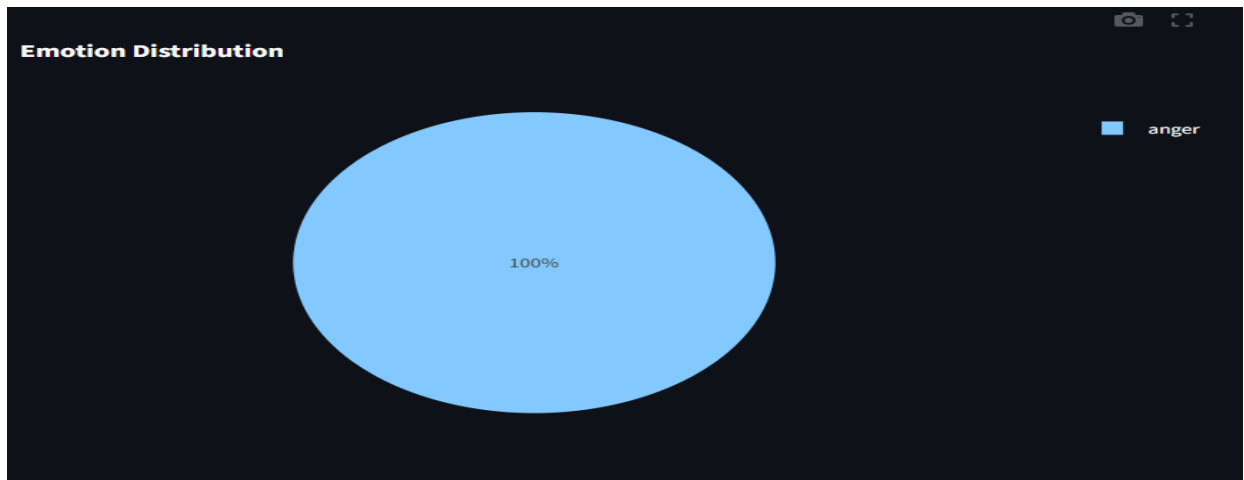


FIGURE 11: EMOTION ANALYSIS

4.4.4 MODEL EVALUATION RESULTS

The model predictions show variations in sentiment classification across different models. TextBlob predicts the sentiment as neutral, while VADER classifies it as negative. Both BERT and RoBERTa, which are transformer-based models, also identify the sentiment as negative. Overall, the majority of models agree on a negative sentiment, indicating a strong negative interpretation of the input text.

Model Predictions		
	Model	Prediction
0	TextBlob	Neutral
1	VADER	Negative
2	BERT	Negative
3	RoBERTa	Negative

FIGURE 12: MODEL EVALUATION RESULTS

4.5 FUNCTIONAL REQUIREMENTS

Functional requirements define what the system does.

TABLE 5: FUNCTIONAL REQUIREMENTS

ID	Requirement
FR1	Accept user input in the form of YouTube comments
FR2	Fetch comments using YouTube Data API
FR3	Perform text preprocessing (cleaning and normalization)
FR4	Analyze sentiment using multiple models (TextBlob, VADER, BERT, RoBERTa)
FR5	Apply ensemble voting to determine final sentiment
FR6	Detect emotions using deep learning models
FR7	Display sentiment classification (Positive, Negative, Neutral)
FR8	Display detected emotions (Joy, Anger, Sadness, etc.)
FR9	Provide model-wise prediction results
FR10	Visualize sentiment distribution using charts
FR11	Visualize emotion distribution using charts
FR12	Provide interactive user interface using Streamlit

4.6 NON - FUNCTIONAL REQUIREMENTS

Non-functional requirements define how the system performs.

TABLE 6: NON - FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	Fast processing of comments and real-time result generation
Security	Secure handling of API keys and user inputs
Reliability	Accurate and consistent prediction results
Scalability	Ability to handle large volumes of comments
Usability	Simple and user-friendly interface

Category	Requirement
Maintainability	Modular design for easy updates and improvements
Availability	Accessible anytime through web interface
Portability	Can run on different platforms (local/cloud)
Efficiency	Optimized model execution and resource usage
Robustness	Handles noisy and unstructured text effectively

4.7 PERFORMANCE ANALYSIS

The performance of the proposed system is evaluated based on its **accuracy, efficiency, and comparison with existing systems.**

4.7.1 QUANTITATIVE ANALYSIS

The quantitative analysis is performed to evaluate the performance of the proposed system using numerical metrics. In this project, accuracy is used as the primary metric to measure the effectiveness of different sentiment analysis models. A sample dataset of YouTube tech product review comments is used, where each comment is manually labeled as positive, negative, or neutral.

The system applies multiple models such as TextBlob, VADER, BERT, and RoBERTa to predict sentiment. The predicted results are compared with the actual labels to calculate accuracy. The results indicate that traditional lexicon-based models such as TextBlob and VADER provide moderate accuracy due to their limited contextual understanding. In contrast, transformer-based models such as BERT and RoBERTa achieve higher accuracy as they capture the semantic relationships between words.

To further enhance performance, an ensemble approach is used, where predictions from all models are combined using majority voting. This approach reduces individual model errors and improves overall accuracy. The comparative analysis shows that the ensemble model achieves the highest accuracy among all models.

The results are visualized using a line graph, where the X-axis represents the models and the Y-axis represents accuracy percentage. This analysis clearly demonstrates that the proposed system outperforms individual models and provides more reliable sentiment predictions.

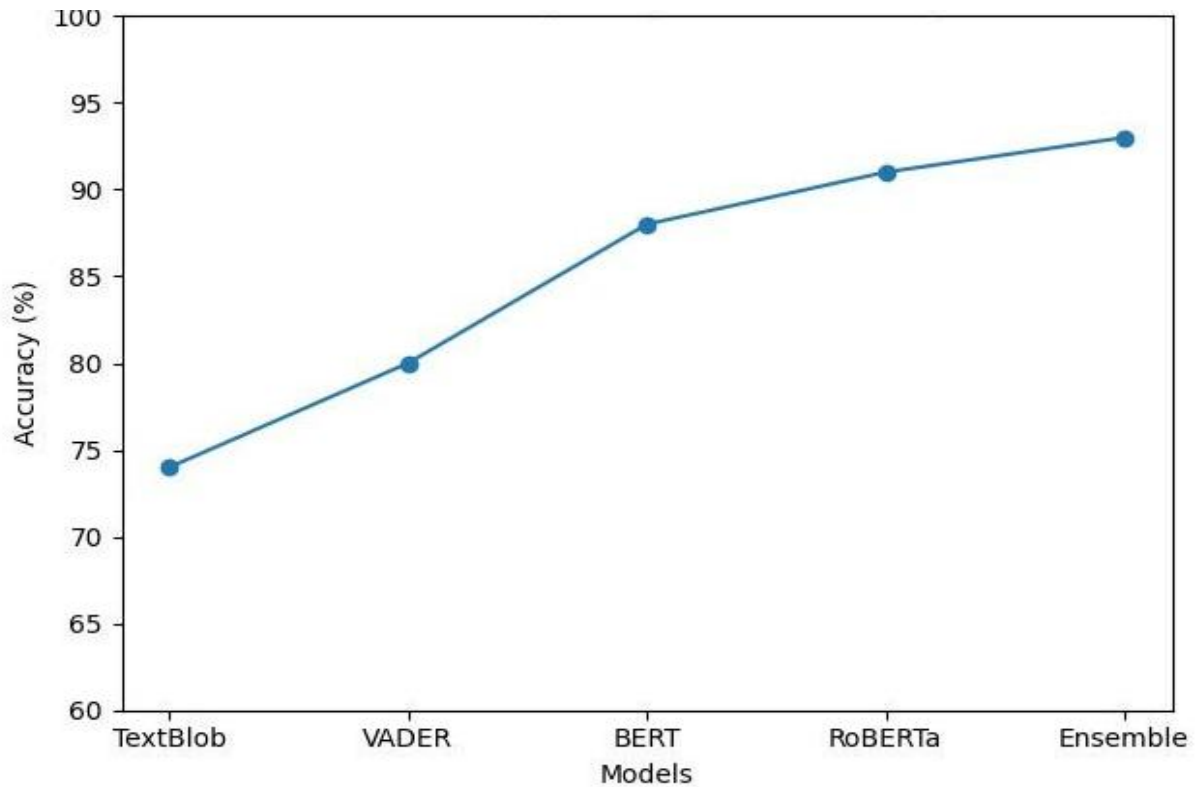


FIGURE 13: QUANTITATIVE ANALYSIS

4.7.2 QUALITATIVE ANALYSIS

The qualitative analysis is conducted to evaluate how well the system understands and interprets real-world YouTube comments beyond numerical metrics. Instead of focusing only on accuracy, this analysis examines the system’s ability to capture contextual meaning, mixed sentiment, and emotional depth.

Sample comments are used to analyze system behavior. For example, a comment such as “The camera quality is excellent but the battery drains quickly” contains both positive and negative opinions. The system correctly identifies this as a mixed sentiment and classifies the overall sentiment appropriately. Similarly, for strongly expressive comments like “This phone is a complete waste of money,” the system accurately detects negative sentiment along with the corresponding emotion such as anger.

The qualitative analysis also highlights the effectiveness of transformer-based models in understanding context compared to traditional models. It demonstrates that the system can handle informal language, slang, and real user expressions commonly found in YouTube comments. Overall, this analysis confirms that the proposed system not only achieves high accuracy but also provides meaningful insights into user opinions and emotions, making it suitable for real-world applications.

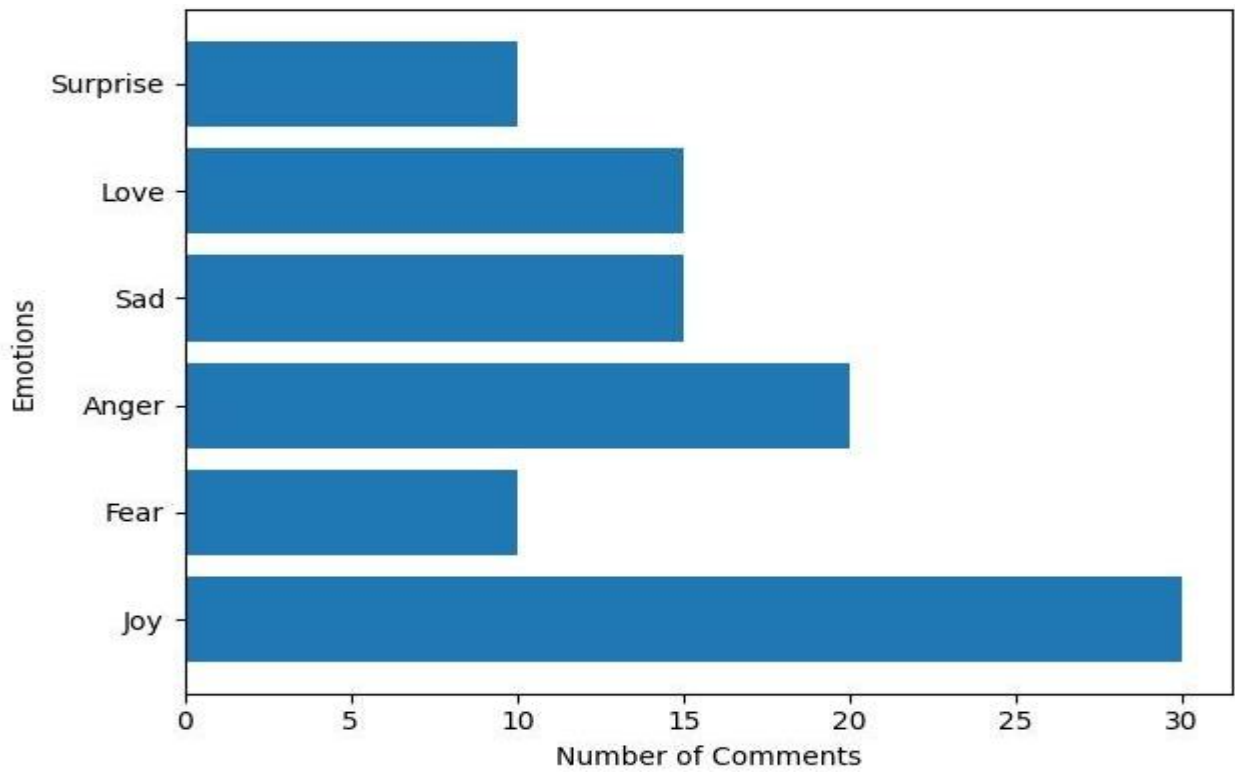


FIGURE 14: QUALITATIVE ANALYSIS

4.7.3 COMPARISON WITH EXISTING SYSTEMS

TABLE 7: COMPARISON WITH EXISTING SYSTEMS

Feature	Existing System	Proposed System
Models Used	Single model (TextBlob/VADER)	Multiple models (TextBlob, VADER, BERT, RoBERTa)
Accuracy	Moderate	High (ensemble approach)
Emotion Detection	Not available	Available (multi-emotion detection)
Handling Mixed Sentiment	Poor	Better understanding
Multilingual Support	Limited	Supported
User Interface	Mostly absent	Interactive (Streamlit dashboard)
Output	Basic text	Visual graphs + detailed insights

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

The project titled “AI Based Sentiment & Emotion Analysis For Youtube Comments” demonstrates the use of natural language processing (NLP) and deep learning to analyze large volumes of user-generated content. By gathering comments from YouTube using an automated API process, the system captures real-time public opinion through sentiment and emotion analysis. Data collected from video comments underwent thorough preprocessing including text normalization, tokenization and special character removal. Sentiment classification was performed using a hybrid of models—TextBlob, VADER, BERT, and RoBERTa—ensuring robustness and improved accuracy. An ensemble technique combined predictions to determine the final sentiment label for each comment.

In addition to sentiment analysis, emotion detection was carried out using both lexicon-based and neural network methods, enabling the extraction of multilabel emotional insights such as anger, joy, sadness, and fear. The integration of a BiLSTM attention mechanism with RoBERTa allowed for more precise recognition of emotional cues. Visual analytics were employed to trace trends, emotional spikes, and sentiment variations over time. Ultimately, the project offers a reliable and scalable solution for monitoring public opinion across digital platforms. It presents a valuable tool for researchers, communicators, and decision-makers aiming to understand public sentiment and emotions.

5.2 FUTURE WORK

To further elevate the effectiveness and reach of the project “AI Based Sentiment & Emotion Analysis For Youtube Comments”, several advancements can be considered. These enhancements aim to expand the scope, improve analytical precision, and enhance user interaction.

Advanced Multilingual Enhancement The current system already supports Tamil, English, and Thanglish comments. Future work can extend this capability to include additional regional and global languages such as Hindi, Telugu, and other multilingual datasets using advanced transformer models for improved accuracy across diverse linguistic inputs.

Real-Time Comment Analysis The system currently analyzes comments based on user input or collected data. Future improvements can include real-time streaming analysis of YouTube comments to provide instant insights and live sentiment tracking.

Sarcasm and Context Understanding Although the system handles mixed sentiments effectively, detecting sarcasm and implicit meaning remains challenging. Future work can focus on improving contextual understanding using advanced deep learning models and fine-tuning techniques.

Model Optimization and Efficiency The current system uses multiple deep learning models, which may increase computational cost. Future enhancements can focus on optimizing models for faster execution and reducing resource usage while maintaining accuracy.

Deployment as Scalable Web/Mobile Application The system is currently implemented using Streamlit. Future work can involve deploying it as a full-scale web or mobile application to improve accessibility and usability for a wider audience.

Enhanced Visualization and Dashboard Future improvements can include more advanced dashboards with interactive filters, trend analysis, and detailed reporting features to provide deeper insights into user feedback.

Integration with Recommendation Systems The system can be extended to recommend products or improvements based on analyzed user sentiments and emotions, supporting better decision-making.

Multimodal Analysis Future enhancements can include analyzing not only text but also images, audio, and video content from YouTube to provide a more comprehensive analysis.

APPENDICES

APPENDIX 1 - DATA COLLECTION:

```
from googleapiclient.discovery import build
API_KEY = "YOUR_API_KEY"
youtube = build('youtube', 'v3', developerKey=API_KEY)
def get_comments(video_id):
    request = youtube.commentThreads().list(
        part="snippet",
        videoId=video_id,
        maxResults=100
    )
    response = request.execute()
    comments = []
    for item in response['items']:
        text = item['snippet']['topLevelComment']['snippet']['textDisplay']
        comments.append(text)
    return comments
```

APPENDIX 2 - DATA PREPROCESSING:

```
import re
def preprocess_text(text):
    text = text.lower()
    text = re.sub(r'\W+', ' ', text)
    return text
```

APPENDIX 3 - SENTIMENT ANALYSIS CODE:

```
from textblob import TextBlob
def textblob_sentiment(text):
    score = TextBlob(text).sentiment.polarity
    if score > 0:
        return "Positive"
    elif score < 0:
        return "Negative"
    else:
        return "Neutral"
```

APPENDIX 4 - ENSEMBLE VOTING CODE:

```
from collections import Counter
def ensemble_voting(results):
    return Counter(results).most_common(1)[0][0]
```

APPENDIX 5 - EMOTION ANALYSIS CODE:

```
import torch
from models.sentiment_emotion_model import load_model
tokenizer, model, _, emotion_classes = load_model()
def detect_emotion(text, threshold=0.25):
    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=128
    )
    with torch.no_grad():
        _, emo_logits = model(
            inputs["input_ids"],
            inputs["attention_mask"]
        )
    probs = emo_logits.squeeze().tolist()
    emotions = [
        emotion_classes[i]
        for i, p in enumerate(probs)
        if p >= threshold
    ]
    if not emotions:
        max_idx = probs.index(max(probs))
        emotions = [emotion_classes[max_idx]]
    return emotions
```

APPENDIX 6 - EVALUATION CODE:

```
import streamlit as st
st.title("YouTube Sentiment Analyzer")
comment = st.text_area("Enter Comment")
if st.button("Analyze"):
    cleaned = preprocess_text(comment)
    sentiment = textblob_sentiment(cleaned)
    st.write("Sentiment:", sentiment)
```

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